

Advanced Applied Econometrics: Static discrete choice labour supply

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Stochastic discrete choice labour supply models are very popular in empirical research and have frequently been used as a tool to perform tax policy analysis. In this problem set we will examine a basic example of this class of model, and for simplicity we will abstract from any demographic heterogeneity (although it is straightforward to incorporate).

Random utility model:

Individuals i choose to work $h_{ij} \in \{0, 10, 20, 30, 40\}$ hours per week, where j indexes the discrete alternatives. Preferences over these alternatives are described by the parametric utility function:

$$U_{ij} = \frac{c_{ij}^\theta}{\theta} - \alpha h_{ij} + \varepsilon_{ij} \quad \forall i, j$$

where ε_{ij} are i.i.d. Type-I extreme value distributed preference shocks, θ is the coefficient of relative risk aversion, and α captures the disutility of working hours. The scale parameter γ is normalised to 1 throughout.

Budget constraint and wage equation:

$$\begin{aligned} c_{ij} &= y_i + \omega_i \cdot h_{ij} - T(\omega_i \cdot h_{ij}) \\ \log(\omega_i) &= \mu_\omega + \varepsilon_i^\omega, \quad \varepsilon_i^\omega \sim \mathcal{N}(0, \sigma_\omega^2) \end{aligned}$$

where y_i is non-labour income, ω_i is the individual gross hourly wage, and $T(\cdot)$ denotes the tax function:

$$T(E) = \begin{cases} 0 & \text{if } E \leq 80 \\ t \cdot (E - 80) & \text{if } E > 80 \end{cases}$$

with marginal tax rate $t = 0.3$ applied to weekly labour earnings $E = \omega_i \cdot h_{ij}$ above the threshold. Non-labour income is not taxed.

Calibrated parameters for simulation:

- Non-labour income: $y_i \sim \text{Uniform}[10, 100]$
- Wage equation: $\mu_\omega = 1, \quad \sigma_\omega = 0.55$
- Preference parameters: $\theta = 0.3, \quad \gamma = 1$ (normalised), $\alpha = 0.1$

Problem 1: General questions

1. Why is a discrete work-hours choice framework attractive for modelling labour supply? Discuss at least three arguments, drawing on (i) the empirical distribution of working hours, (ii) the structure of tax-transfer systems, and (iii) computational considerations.
2. The choice set used here is $\{0, 10, 20, 30, 40\}$ hours per week. What considerations should guide the specification of the choice set in practice? What are the potential costs of including too few or too many alternatives?
3. For an individual with non-labour income $y_i = 50$ and hourly wage $\omega_i = 6$, compute consumption c_{ij} at each hours alternative and sketch the resulting budget set (consumption on the vertical axis, hours on the horizontal). At which point does the tax start to bind, and what does the kink look like?
4. Discuss the role of each preference parameter (θ, α, γ) in the utility function.
5. What does this model abstract from? List at least three dimensions and, for each, briefly indicate how the omission may affect policy conclusions.

Problem 2: Simulating a dataset

1. Using the data generating process above, simulate a cross-sectional dataset of $N = 10,000$ individuals comprising each individual's hours worked h_i , non-labour income y_i , labour earnings $\omega_i \cdot h_i$, and hourly wage ω_i . Note that wages are observed only for workers ($h_i > 0$).
2. Plot the distribution of work hours across the five alternatives. How do average non-labour income and average wages vary across hours categories?
3. Compute the tax revenue.

Problem 3: Estimation of preference parameters

1. Write down the log-likelihood function to estimate (α, θ) . Note that $\gamma = 1$ is normalised and not estimated. What problem arises for non-workers, and how would you address it?
2. Implement the log-likelihood and estimate the model using the simulated dataset. To keep the exercise simple and well identified, fix $\theta = 0.3$ (its true value) and $\gamma = 1$, and estimate only the disutility parameter α . Report the estimated α and verify that it is close to the true value used for simulation.
3. Compare observed and predicted hours-choice frequencies following van Soest (1995, Table 3). A validated fit justifies using the estimated parameters for the counterfactual exercise in Problem 4.¹

Problem 4: Counterfactual policy simulation A key advantage of structural estimation is that the recovered parameters can be used to simulate counterfactual outcomes under alternative policies, holding individual characteristics fixed. Using the preference parameters estimated in Problem 3:

1. The government is considering lowering the tax-free earnings threshold from €80 to €50 per week, keeping the marginal tax rate at $t = 0.3$. Simulate the behavioural response by applying this reform to the individuals in the baseline dataset, i.e. holding the calibration wages, non-labour income, and taste shocks fixed.

¹Because the data are generated from the same parametric model, the homogeneous model fits the simulated data well here. Problem 5 revisits this comparison on a dataset with unobserved type heterogeneity, where the homogeneous model falls short and motivates the finite-mixture extension.

2. Plot and compare the distribution of hours worked under the reform against the baseline from Problem 2.
3. Compute tax revenue under the reform. Decompose the change relative to the baseline into a *mechanical effect* (revenue change at unchanged behaviour) and a *behavioural effect* (additional revenue change due to labour supply adjustment).

Problem 5: Unobserved individual heterogeneity Observed heterogeneity (e.g. education, experience) can be incorporated directly into the wage equation. Structural models can additionally capture *unobserved* time-invariant heterogeneity using a finite mixture approach. In preparation for the lecture, you have read Keane and Wolpin (1997); re-examine p. 481f, where unobserved endowment types are introduced.

1. Suppose individuals differ in their time-invariant disutility of work. There are K unobserved types, each characterised by a type-specific parameter α_k and population share π_k , with $\sum_k \pi_k = 1$. The remaining preference parameter θ is common across types ($\gamma = 1$ is normalised throughout). Write down the mixture log-likelihood function.
2. Assume two types $k \in \{1, 2\}$ with $\alpha_1 = 0.02$, $\alpha_2 = 0.40$, $\pi_1 = 0.5$, $\pi_2 = 0.5$, and $\gamma = 1$ (normalised), $\theta = 0.3$ as before. Simulate a dataset of $N = 10,000$ individuals by first drawing each individual's type, then simulating hours choices using the corresponding α_k .

3. Model fit comparison.

- As a baseline, estimate the *homogeneous* model from Problem 3 on the heterogeneous dataset simulated in 5.2. As in Problem 3, fix $\theta = 0.3$ and $\gamma = 1$ and estimate only a single pooled α . Following van Soest (1995, Table 3), produce a table comparing observed and model-predicted choice frequencies at each hours alternative. Is the fit adequate?
- Now estimate the mixture model on the same data, with θ fixed at its true value $\theta = 0.3$ and $\gamma = 1$; estimate only the type-specific disutilities and the mixing share $(\alpha_1, \alpha_2, \pi_1)$. Reproduce the fit table and compare it to the homogeneous baseline. Does the additional flexibility improve fit? Report the estimated parameters $(\alpha_1, \alpha_2, \pi_1)$ and verify that they are close to the true values.